

Computability: Exercise 1

Due: April 11, 2026

Read these notes first. As usual in theoretical courses, a complete answer to a question must include a proof of correctness unless the question specifically states otherwise. For questions that require constructing a specific automaton for a language L , however, you are not expected to give a full proof by induction unless specifically requested by the question. Instead, give an explanation that would be sufficient for an experienced person to be convinced that you understand why the automaton does what you say it does.

Also, note that some questions below contain definitions of some concepts in a format such as

Definition (concept). *A definition of the concept.*

These definitions are important, and you are expected to learn, know and remember them. Also, a definition that appears in one question may be used without reference in questions that appear below it.

Question 1

Let $\Sigma = \{a, b\}$. For each of the following languages, draw a DFA that decides it. Justify your answer briefly by explaining, for every state q , which words in Σ^* end in q .

- Σ^* (namely the language that contains all words)
- $\emptyset$
- $\{\epsilon\}$
- In this section you should draw a DFA over the alphabet $\Psi \stackrel{\text{def}}{=} \{0, 1\}$ rather than Σ . The language is comprised of all words that, when considered as a number in base 2, are divisible by 3. Consider the empty word as representing the number zero.

Hint: The DFA should include a state for each possible remainder when dividing by 3.

Question 2

Let $\Sigma = \{a, b\}$, and define the following language:

$$L \stackrel{\text{def}}{=} \{w \in \Sigma^* : w \text{ contains the sequence } bba\}$$

- Draw a DFA $\mathcal{A}$ that decides L .
- Give a full formal definition of $\mathcal{A}$.
- Give a full formal proof (by induction) that $L(\mathcal{A}) = L$.

Question 3

Definition (concatenation of words). For an alphabet Σ and $u, v \in \Sigma^*$, we define the concatenation of u and v by

$$u \cdot v \stackrel{\text{def}}{=} u_1 u_2 \dots u_n v_1 v_2 \dots v_m,$$

where we write $u = u_1 u_2 \dots u_n$ and $v = v_1 v_2 \dots v_m$. We often denote the concatenation by uv , omitting the dot.

Note that to make sure that concatenation is well defined, we must denote words by letters that do not belong to the alphabet. So for example over the alphabet $\{a, b, \#\}$, if we set $v = ab$ and $w = bba$ then $vw = abw = vbba = abbbba$. Similarly, $v\#w$ is a concatenation of three words, one of which is the single-letter word $\#$, and we have $v\#w = ab\#bba$.

Let $L_1, L_2 \subseteq \Sigma^*$ be languages over $\Sigma = \{0, 1\}$. Define a larger alphabet $\Sigma' = \{0, 1, \#\}$, and let

$$L \stackrel{\text{def}}{=} \{u\#v : u \in L_1, v \in L_2\}$$

Prove that if L_1 and L_2 are regular, then so is L .

Question 4

Definition (concatenation of languages). For languages $L_1, L_2 \subseteq \Sigma^*$, we define their concatenation by $L_1 \cdot L_2 \stackrel{\text{def}}{=} \{uv : u \in L_1, v \in L_2\}$.

Let $\Sigma = \{a, b\}$, and define the following languages:

$$L = \{w \in \Sigma^* : w \text{ ends in } bba\}, \quad L' = \{w \in \Sigma^* : w \text{ ends in } b\}$$

For each of the following languages, show a DFA that decides it.

- a. $L \cdot L'$
- b. $L \cdot L$

Hint: First understand each of the above languages.

Question 5

Given two alphabets Σ_1, Σ_2 , and a function $f : \Sigma_1 \rightarrow \Sigma_2^*$ that maps letters in the first alphabet to words in the second alphabet, we define the function $\hat{f} : \Sigma_1^* \rightarrow \Sigma_2^*$ that maps words in the first alphabet to words in the second alphabet as follows:

$$\hat{f}(w_1 \dots w_n) = f(w_1) \cdot \dots \cdot f(w_n),$$

that is, $\hat{f}$ uses f to map the letters to words and concatenates them.

Given a language $L_2 \subseteq \Sigma_2^*$ and a DFA $\mathcal{A}_2 = \langle Q_2, \Sigma_2, \delta_2, q_0^2, F_2 \rangle$ that decides L_2 , construct a DFA $\mathcal{A}_1 = \langle Q_1, \Sigma_1, \delta_1, q_0^1, F_1 \rangle$ that decides L_1 where

$$L_1 := \{w \in \Sigma_1^* : \hat{f}(w) \in L_2\}.$$

Question 6

Let $\mathcal{A} = \langle Q, \Sigma, \delta, q_0, F \rangle$ be a DFA, and let $L = L(\mathcal{A})$ be the language decided by $\mathcal{A}$.

We define the even prefix closure of L as:

$$\text{Prefix}(L) := \{w \in \Sigma^* \mid \exists w' \in \Sigma^* \text{ such that } w \cdot w' \in L\}.$$

That is, $\text{Prefix}(L)$ consists of all words that are prefixes of some word in L .

Prove that there exists a DFA that decides $\text{Prefix}(L)$.

Question 7 (Optional)

Let Σ be an alphabet. In this question, we will show that there are more languages $L \subseteq \Sigma^*$ than words $w \in \Sigma^*$.

- a. Show that you can order all words $w \in \Sigma^*$.

(That is, you can list the words as $w_1, w_2, w_3, \dots \in \Sigma^*$.)

- b. Prove that you cannot order all languages $L \subseteq \Sigma^*$.

Hint: Assume, for contradiction, that you can enumerate the languages as $L_1, L_2, L_3, \dots \subseteq \Sigma^*$. Consider the language L consists of words:

$$w_n \in L \iff w_n \notin L_n$$

and derive a contradiction.